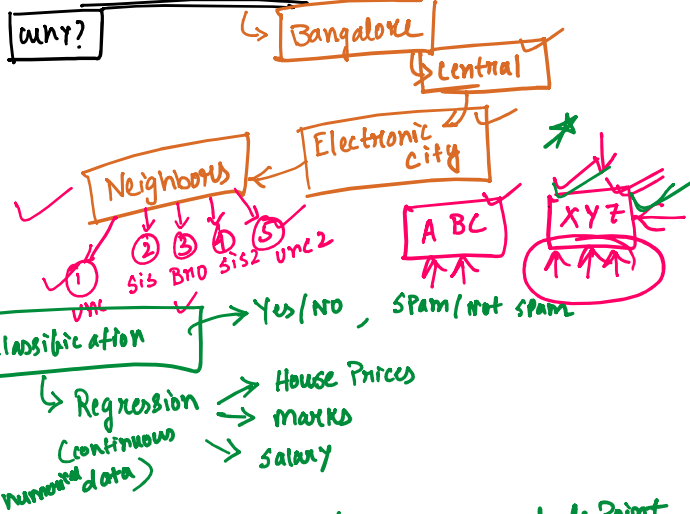


KNN $\rightarrow$ K - Nearest Neighbors



$\rightarrow$ Predicting the answer from a new data point, by looking at the most similar existing data points.

KNN $\rightarrow$ K - Nearest Neighbors
 $\rightarrow$ No. of neighbors

$\rightarrow$ with help of which $\uparrow$ will be predicting the answer for new data point.

Types of KNN:-

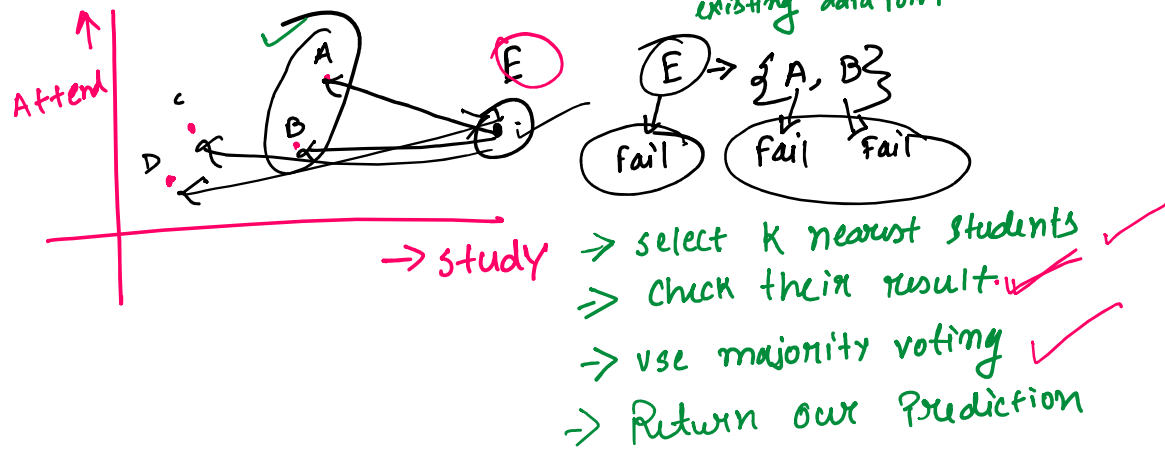
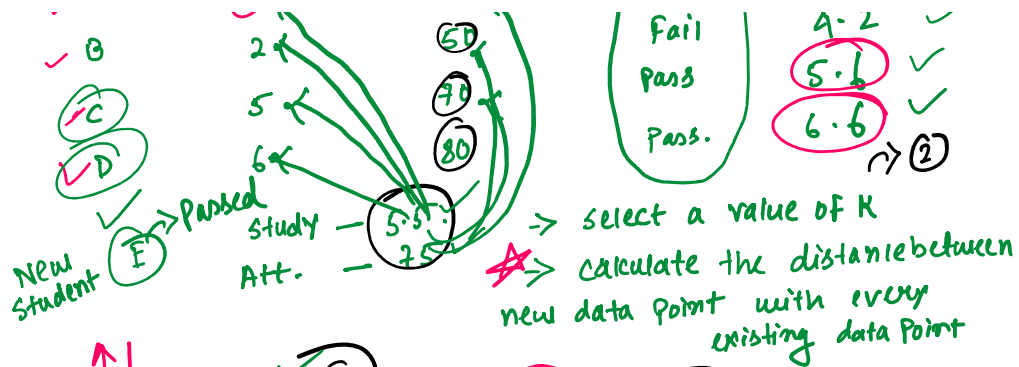
1) Classification
 $\rightarrow$ Spam/Not Spam
 $\rightarrow$ Passed/Failed
 (Categorical)

2) Regression
 $\rightarrow$ Predict House Price
 $\rightarrow$ Salary.
 (Numerical)

Why? $\rightarrow$ KNN

1) How KNN Classification Algorithm works?

	Study Hours	Attendance	Result	Score (CGPA)
✓ A	1	40	Fail	3.2 ✓
✓ B	2	60	Fail	4.2 ✓
	3	70	Pass	5.1 ✓



2) How KNN Regression Algorithm works?

C, D ⇒ 5.6, 6.6

$$\hat{E} = \frac{5.6 + 6.6}{2}$$

To calculate distance from new data point to every existing Data Point:—

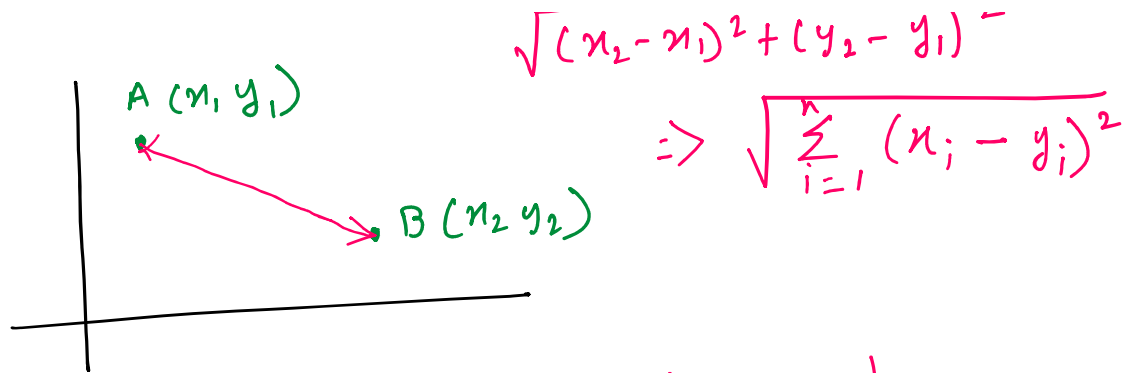
1) Euclidean Distance

2) Manhattan Distance.

$$A(x_1, y_1)$$

$$d(A, B) =$$

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



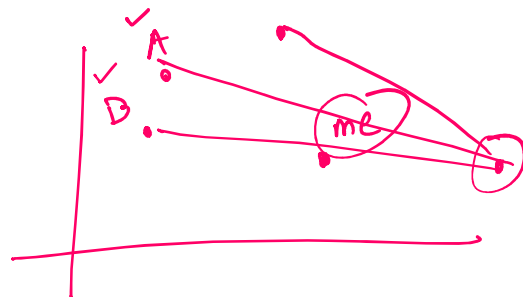
$$d(A, B) \Rightarrow |x_2 - x_1| + |y_2 - y_1|$$

KNN $\Rightarrow$ Lazy learner.

✓ [300 rows] new

KNN $\Rightarrow$ K - nearest neighbors.
 $\hookrightarrow$ "how many nearest"

very small K :-
 $\rightarrow$ Sensitive to distractions.



✓ South
 ✓ ABC

✓ North
 XYZ

K = 100

$\rightarrow$ Sensitive noise

→ overfit

Best way to choose value:-

$$K = \sqrt{n}$$

$$K = \sqrt{n} = 10$$

$\therefore n = \text{No. of samples.}$

Note :- way only for starting point.

weighted KNN

$$K = 3$$

distances

0.1

0.8

0.7

Predict

Fail

Pass

pass.

Pass

Predict = Pass

weighted distance:

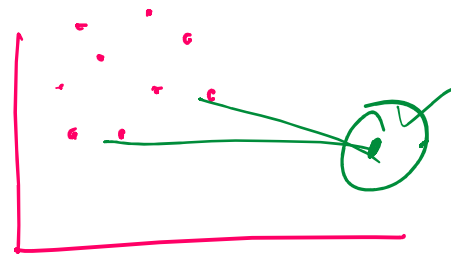
$$w = \frac{1}{d + \epsilon}$$

↓
Distance of nearest neighbor

Why feature scaling is important.

Why feature scaling is important.

	<u>Age</u>	<u>Salary</u>
A	25 ✓	30000
B	26 ,	80000



Manhattan

$$|x_2 - x_1| + |y_2 - y_1|$$

$$|26 - 25| + |80000 - 30000|$$

$$1 + 50000$$

model Parameter Tuning

n-neighbors $\rightarrow$ value of k

$\sqrt{n}$
 $\rightarrow$ CV technique.

weights = "uniform" / "distances"

metric = "manhattan", / "Euclidean"

$\Rightarrow$
 $\Rightarrow$

$\rightarrow$ Normally distributed
 $\rightarrow$ not having much outliers.
 $\rightarrow$ straight line.

→

→ Not having much variance
→ Straight line.

algorithm = {
"brute" ✓
"kd-tree" ✓
"ball-tree" ✓
"auto" ✓

Brute Force



→ Dataset small
→ No. of columns is large.

KD-Tree

Divide the feature space into regions to search faster.

Ball-Tree

How to select the best value of k ?

→ $k = \sqrt{n}$

$\rightarrow K = \sqrt{n}$ $\hookrightarrow$ No. of samples.

Suppose

150 rows of data.

$\hookrightarrow$ 120 training 30 testing

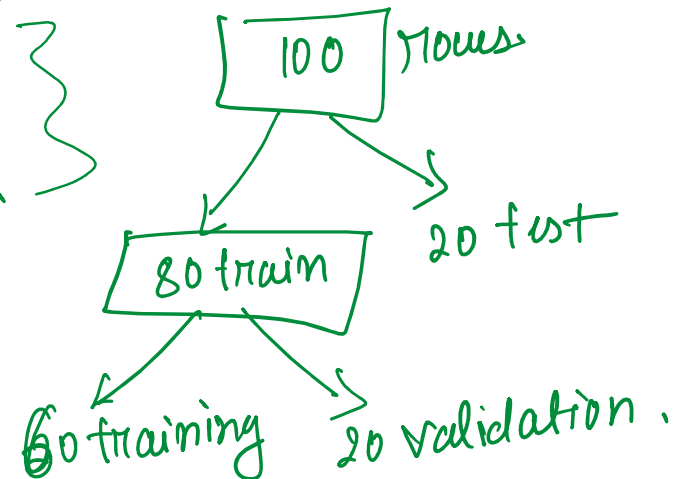
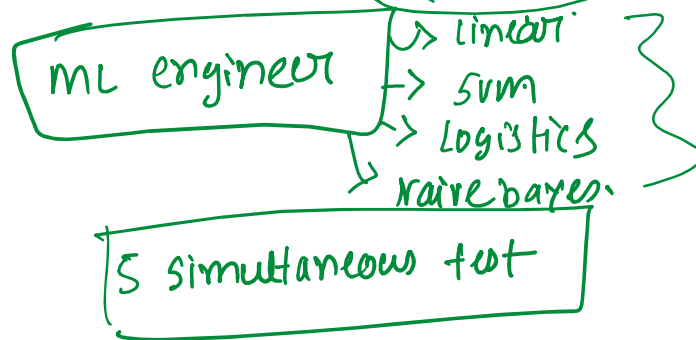
$K = 1, 2, 3, \dots$

Cross Validation

Taking the training dataset.

$\hookrightarrow$ Dividing training dataset into multiple parts.

Training vs testing vs validation $\hookrightarrow$ Train each parts
 $\hookrightarrow$ validate each part.



$k = 1, 2, 3 \dots$

⑤ fold cross validation. →

$x_{\text{train}} = \frac{120 \text{ rows}}{5 \text{ groups}}$

5 Rounds of training
5 ^{random} groups of data.

Fold 1 → 24 rows.

Fold 2 → 24 rows

⋮

Round 1

Fold 1 → Validation

Fold 2, 3, 4, 5 → Training → 95% (accuracy)

$k = 2$

Round 2

Fold 2 → Validation

Fold 1, 3, 4, 5 → Training

→ accuracy 96%

95 + 96 + 92 + 98 + 94

$\sqrt{95\%}$

200 $\sqrt{200}$ $k = \sqrt{n}$

$$K = \boxed{17, 16} \quad \boxed{95\%}$$

for K in range $(1, \underline{16})$:

DF

<u>Age</u>	<u>Gender</u>	<u>BMI</u>	<u>Cancer</u>
23	M	102	Y
24	F	62	N
22	M	56	Y

$$dh = \left\{ \text{'Age': } [23, 24, 22], \text{'Gender': } [\text{'M'}, \text{'F'}, \text{'F'}] \right\}$$